

*?? **Fun Fact:** During the historic Bitcoin crash of 2018, the RSI indicator on the weekly timeframe dropped below 30 for the first time since 2015. Traders who recognized this rare oversold signal and purchased Bitcoin during this period (around \$3,500) saw their investment increase nearly fifteen-fold by 2021 when Bitcoin reached \$64,000. While many investors were caught in the emotional panic of the crash, technical traders focused on the RSI had a quantifiable reason to consider the market extremely oversold, highlighting how technical indicators can sometimes help traders remain rational during highly emotional market periods.*

Bollinger Bands: Measuring Volatility

Created by John Bollinger in the early 1980s, Bollinger Bands consist of three lines: a middle band (typically a 20-day SMA) and upper and lower bands placed two standard deviations away from the middle band. These bands expand and contract based on market volatility, making them a versatile tool for traders.

Imagine Bollinger Bands as the boundaries of a river. During calm periods, the river flows within narrow banks (low volatility). During floods, the river expands beyond its normal boundaries (high volatility). Traders watch for prices approaching or exceeding these "banks" as potential signals.

```
1. def calculate_bollinger_bands(data, window=20, num_std=2):
2.     # Calculate middle band (SMA)
3.     middle_band = data.rolling(window=window).mean()
4.
5.     # Calculate standard deviation
6.     rolling_std = data.rolling(window=window).std()
7.
8.     # Calculate upper and lower bands
9.     upper_band = middle_band + (rolling_std * num_std)
10.    lower_band = middle_band - (rolling_std * num_std)
11.
12.    return middle_band, upper_band, lower_band
13.
14. # Calculate Bollinger Bands
15. apple['BB_Middle'], apple['BB_Upper'], apple['BB_Lower'] =
calculate_bollinger_bands(apple['Close'])
```